

Cataract Detection using CNN & Data Structures

1. Introduction

Cataract is a common eye disease that causes clouding of the lens, leading to vision loss if untreated. Early detection is crucial for preventing blindness.

In this project, we developed an AI-based system using **Convolutional Neural Networks (CNN)** to classify eye images as **Normal** or **Cataract**.

We also integrated **Data Structures (Hash Table & Queue)** for efficient storage and retrieval of predictions.

2. Objectives

- Automate cataract detection using deep learning.
 - Build a user-friendly web application with **Streamlit**.
 - Demonstrate the use of **Hash Table (for mapping image → prediction)** and **Queue (for recent results)**.
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3. Methodology

1. Dataset Preparation

- Collected labeled images (Normal vs Cataract).
- Preprocessed images (resized to 150×150, normalized).

2. Model Training (CNN)

- Convolutional + Pooling layers for feature extraction.
- Dense layers for classification.
- Output: Binary (Normal / Cataract).

3. Web Application (Streamlit)

- Upload image → Preprocessing → Prediction → Display result.

4. Data Structures in Action

- **Hash Table:** Stores results (filename → prediction).
- **Queue:** Maintains the last 10 predictions in order (FIFO).

4. Implementation

- Programming Language: **Python**
- Frameworks/Libraries:
 - TensorFlow/Keras (CNN)
 - Streamlit (Web App)
 - Collections (Queue)
 - JSON (Hash Table visualization)

5. Results

- Achieved **high accuracy** in cataract detection.
- The web app provides real-time predictions with simple drag-and-drop.

- Data Structures (Hash Table + Queue) effectively demonstrate storage and retrieval.
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6. Screenshots

(Add screenshots of your app: upload box, prediction result, hash table, queue)

7. Conclusion

The project successfully demonstrates how **AI + Data Structures** can be combined for healthcare applications.

This system is scalable, easy to use, and can assist doctors in early cataract screening.

8. Future Work

- Extend model to detect multiple eye diseases.
 - Deploy app on cloud for wider accessibility.
 - Improve dataset diversity for higher accuracy.
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9. References

1. TensorFlow Documentation – <https://www.tensorflow.org>
 2. Streamlit Documentation – <https://streamlit.io>
 3. Ophthalmology Research Papers on Cataract Detection
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10. Developer



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